Check for updates

Hammill Institute on Disabilities

Focus on Autism and Other Developmental Disabilities
1–5
© Hammill Institute on Disabilities 2018
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/1088357618794911
focus.sagepub.com
SAGE

Brief Report: The Effects of a Weighted Blanket on Engagement for a Student With ASD

Kathleen N. Zimmerman, PhD, BCBA[1], Jennifer R. Ledford, PhD, BCBA-D[1], and Katherine E. Severini, MEd, BCBA[1]

Abstract

The effectiveness of a weighted blanket intervention and structured work boxes for improving engagement for a child with autism during math circle time in a general education kindergarten classroom were evaluated relative to baseline using an alternating treatments design. The structured work boxes resulted in higher levels of engagement, and the weighted blanket resulted in levels that were similar to or lower than baseline. Additional research is needed on sensory-based interventions; in the meantime, teachers should use evidence-based practices.

Keywords

weighted blanket, autism spectrum disorders, structured work boxes

Stereotypy has been associated with negative outcomes (e.g., decreased engagement) for children with autism spectrum disorders (ASD; Morrison & Rosales-Ruiz, 1997). It is often assumed to be automatically maintained and the result of sensory differences exhibited by children with ASD (Rogers, Hepburn, & Wehner, 2003). Sensory-based interventions (SBIs) are often prescribed to reduce interference of stereotypy with functional and academic activities; parents of children with ASD report SBIs are among the most commonly used treatments for their children (Goin-Kochel, Mackintosh, & Myers, 2009; Hanson et al., 2007). However, research suggests that SBIs, such as weighted vests and alternative seating, have insufficient evidence for use (Barton, Reichow, Schnitz, Smith, & Sherlock, 2015).

Weighted blankets are similar to weighted vests in their purported processes and effects, and are often prescribed for use during typical school activities. They may also be utilized to improve sleep behaviors for children with ASD, although they have been shown to be ineffective for that purpose (Gringras et al., 2014). One study to date has specifically assessed the use of weighted blankets for improving engagement in school activities; student behavior did not improve and in two cases deteriorated when the weighted blanket was used during one-on-one instructional activities (Losinski, Cook, Hirsch, & Sanders, 2017). Thus, this intervention may fall under the category of "fad treatments," described by Zane, Davis, and Rosswurm (2008) as

> interventions that use scientific jargon, sound logical . . . and are discussed in the media and on the Internet, where many parents can be exposed to them . . . [but] have no substantial body of research showing that they are effective . . . (p. 46).

Despite their lack of research, some have considered SBIs to be "conventional" (rather than alternative) therapies due to their widespread use in schools (Hanson et al., 2007). Because they are often used and prescribed in nonrefereed publications (e.g., websites), weighted blankets may be perceived as a potential evidence-based practice (EBP) by parents and practitioners. In turn, they may use these interventions, rather than ones that may require more response effort, even if the conventional interventions have been designated as EBPs (e.g., use of structured work boxes [SWB], an EBP for improving academic, social, communication, and challenging behaviors for young children with ASD; Bennett, Reichow, & Wolery, 2011; Wong et al., 2014). The initial research question was regarding whether a SWB intervention would improve engagement relative to a baseline intervention; however, immediately prior to intervention implementation, the teacher asked researchers to evaluate the weighted blanket and SWB as a treatment package. After consulting with the teacher, researchers decided to separately assess SWB and the weighted blanket. Thus the research question for this study was

[1]Vanderbilt University, Nashville, TN, USA

Corresponding Author:
Kathleen N. Zimmerman, The University of Kansas, 1122 West Campus Road, JRP 521, Lawrence, KS 66045, USA.
Email: kathleen.zimmerman@ku.edu

Research Question 1: Will a weighted blanket intervention or a SWB intervention result in improved engagement in a large group activity when compared with a baseline condition, for a child with ASD in a general education kindergarten classroom?

Method

Participant and Setting

Anwar, a 5-year-old Middle Eastern male with a medical diagnosis of ASD was nominated by his general education kindergarten teacher for participation in the study due to low engagement during math circle time activities. The teacher reported Anwar was above grade level in reading and met grade level standards in math, but frequently engaged in stereotypy during math whole-group instruction that prohibited his participation in instructional activities. Compared with a normative sample of age-matched peers, Anwar's teacher reported he had below average communication and overall engagement skills on the *Social Skills Improvement System* (SSIS; Gresham & Elliott, 2008). Using the SSIS, the teacher also rated Anwar as engaging in average levels of cooperation and externalizing/internalizing behaviors for a 5-year-old male, but above average levels of hyperactivity and inattention. Anwar displayed symptomology associated with ASD (e.g., repetitive behaviors, rigidity to changes in routine, nonfunctional routines or rituals) at average levels commensurate with individuals with ASD using the SSIS Autism Spectrum scale (Gresham & Elliott, 2008). He did not receive special education services during math and was being evaluated to receive occupational therapy at the time of the study. The first author, a graduate student in special education, served as the implementer for most sessions; the third author, a graduate student in special education, implemented sessions when the first author was not available.

All sessions were conducted in a general education kindergarten classroom with 20 students and one general education teacher during math circle time on a 1.8 m × 2.7 m carpet. A special education aide was present during the counting portion of each lesson. Math circle time included the following: oral counting in English and Spanish, content instruction with teacher manipulated instructional materials (e.g., shapes or counters), and student opportunities to answer questions using the instructional materials. On average, five to seven students were provided an opportunity to answer questions each lesson. Sessions occurred for the first 10 min of math circle time.

Data Collection

The primary dependent variable was the percentage of intervals in which Anwar was engaged in instruction. Engagement was defined as (a) appropriately manipulating assigned instructional materials (e.g., hundreds chart, shapes, file folder pieces, iPad math application), (b) appropriately communicating with a peer or adult at a designed time (i.e., when asked a question or participating in a group discussion), (c) appropriately transitioning between locations (e.g., from his designated carpet space to a place in a group circle), and (d) orienting toward instruction or instructional materials (e.g., teacher, shared group materials, or file folder game during instruction). Unengagement was defined as (a) eloping from the carpet, (b) failing to manipulate a material or orient to instruction (e.g., sitting on the carpet, but not using materials or orienting to the teacher), or (c) engaging in stereotypy. Stereotypy was defined via teacher report and researcher observation as (a) placing finger(s) across the plane of the lips or entrance to the nostril, (b) pressing fingers together with arms extended to the side, (c) repeating vowel sounds or words (e.g., *cat party, /aaaaaaaa/*), and (d) holding a material in the air parallel to the eyeball with or without repetitively moving the item up and down. Engagement and unengagement, including stereotypy, were mutually exclusive; Anwar's stereotypy was incompatible with engagement behaviors. In addition, the weighted blanket was not a math instructional material; thus, orienting to the blanket or manipulating the blanket was coded as an unengaged behavior. Intervals were denoted as uncodable if Anwar was on the carpet, but his face was not in the frame; this occurred on average for 6.7% (n = 4.3) of intervals per session; the proportion of uncodable intervals was not differential across conditions.

Engagement behaviors were measured via video with ProCoderDV (Tapp, 2003) using momentary time sampling with 10 s intervals. The total percentage of codable intervals with engagement behaviors was calculated by subtracting the number of uncodable intervals from the total number of intervals, then dividing the total number of engaged intervals by the total number of codable intervals and multiplying the quotient by 100.

Interobserver agreement and procedural fidelity data were collected in 33% to 60% of sessions across conditions using point-by-point agreement and direct observational recording (Ayres & Ledford, 2014). Mean agreement was 88.8% (range across sessions: 78%–95%) and mean fidelity was 99.5% (range across sessions: 95%–100%). The presence or absence of critical steps (e.g., absence of reinforcement or verbal prompts, placement of the weighted blanket, presence of the SWB materials) were measured across all conditions using direct observational recording to account for the occurrence of behaviors more than once per session (e.g., placement of the blanket on Anwar's lap at the start of an activity if needed).

Materials and Procedures

All sessions began with the teacher directive "it's time for math" and ended after 10 min of instructional time elapsed. Across all conditions, the teacher conducted the math circle

time lesson consisting of three parts: counting, instruction, and student participation activities. The researcher did not provide reinforcement, task directions, or prompts to maintain engagement *in any condition*. The researcher stood at least 1 m away from Anwar across all conditions. The study was designed without prompts to continue engagement or planned reinforcement to mirror what the general education teacher reported she would be able to continue after the study.

Baseline. During baseline conditions, the researcher recorded the session and told the teacher to conduct the lesson as planned. Anwar was provided one opportunity to respond during each lesson per the typical teacher behavior prior to the study.

Weighted blanket. The general education teacher asked researchers to evaluate the effectiveness of the weighted blanket. A 0.3 m × 0.3 m blanket weighing 1.49 kg (9.2% of Anwar's body weight) was made of cloth fabric and contained nine stitched squares each containing rice that had been soaked with a lavender scented product. Procedures mirrored baseline procedures except the researcher placed the weighted blanket on Anwar before each of the three activities: counting, instruction, and student participation opportunities. After placing the blanket on Anwar's lap, the researcher returned to the 1 m distance maintained across all conditions. If Anwar removed the blanket, the researcher did not prompt blanket use until initiation of the next activity.

SWB intervention. Three rectangular boxes measuring 30.48 cm long and 15.24 cm wide were placed adjacent to Anwar's assigned carpet spot. Each box contained one activity: a hundreds chart on letter-sized paper to allow for individual opportunities to motorically respond while the class engaged in oral counting (a behavior incompatible with his most common stereotypy), a file folder game with eight (counting to 10) or 16 (sorting 2-D shapes) pieces that matched the content instruction in the classroom, and a number handwriting application on an iPad with headphones (chosen based on a related Individualized Education Program [IEP] goal). The researcher used graduated guidance (physical prompts and pointing) to initiate each task, as needed, then returned to the 1 m distance maintained across all conditions. The researcher provided error correction (physical prompts) for the first session for each activity (counting to 10 and sorting 2-D shapes) to teach Anwar how to use the SWB materials; placement of four game items on average were prompted across all sessions (range 3–7).

Experimental Design

An alternating treatments single case research design (Barlow & Hayes, 1979) was used to assess each intervention condition relative to the other *and* to a baseline condition. Conditions were randomly ordered and the presence of a functional relation was evaluated by assessing differences in level and overlapping data between conditions. Initial baseline and best alone conditions were included to detect potential multitreatment interference (differences in behavior potentially related to the rapid alternation between conditions rather than true condition differences).

Results

Engagement data across conditions are displayed in Figure 1. The SWB intervention resulted in consistently higher levels of engagement than the weighted blanket or baseline conditions (80%–100% engagement). The weighted blanket intervention resulted in counter-therapeutic levels of engagement relative to the baseline condition (2%–16% of intervals), with a high degree of overlap. Although there was a decreasing trend in engagement in the SWB condition, data were consistently higher than in baseline or weighted blanket conditions; data stabilized during the best alone condition.

Discussion

Similar to previous research on weighted blankets (Losinski et al., 2017) and other SBIs (Barton et al., 2015), results show the weighted blanket did not result in therapeutic behavior change; in fact, student engagement was lower when compared with baseline. Anecdotally, Anwar was seated on the carpet more often when using the blanket than during baseline sessions, but often engaged in behaviors that were nonfunctional (touching the blanket) or incompatible with engagement in academic responding (e.g., putting blanket on his head). The SWB intervention, an EBP (Wong et al., 2014), provided Anwar with more opportunities to respond with individualized materials as compared with the other conditions, likely resulting in the observed increased engagement.

Broadening opportunities to engage nonverbally may artificially inflate engagement, although nonverbal engagement could have also been coded in baseline or weighted blanket conditions (e.g., if Anwar counted on his fingers along with peer verbal counting). Future research should compare weighted blanket interventions with other interventions, while holding opportunities to respond constant. Given increased opportunities to nonverbally respond is an integral part of the SWB intervention, this confound may be unimportant. However, because the SWB intervention provided active response opportunities and the weighted blanket intervention was devised without such (in keeping with use and purported sensory effects), it is possible that the most relevant comparisons are SWB compared with baseline and weighted blanket compared with baseline. An additional condition (SWB + weighted blanket) would have allowed us to detect

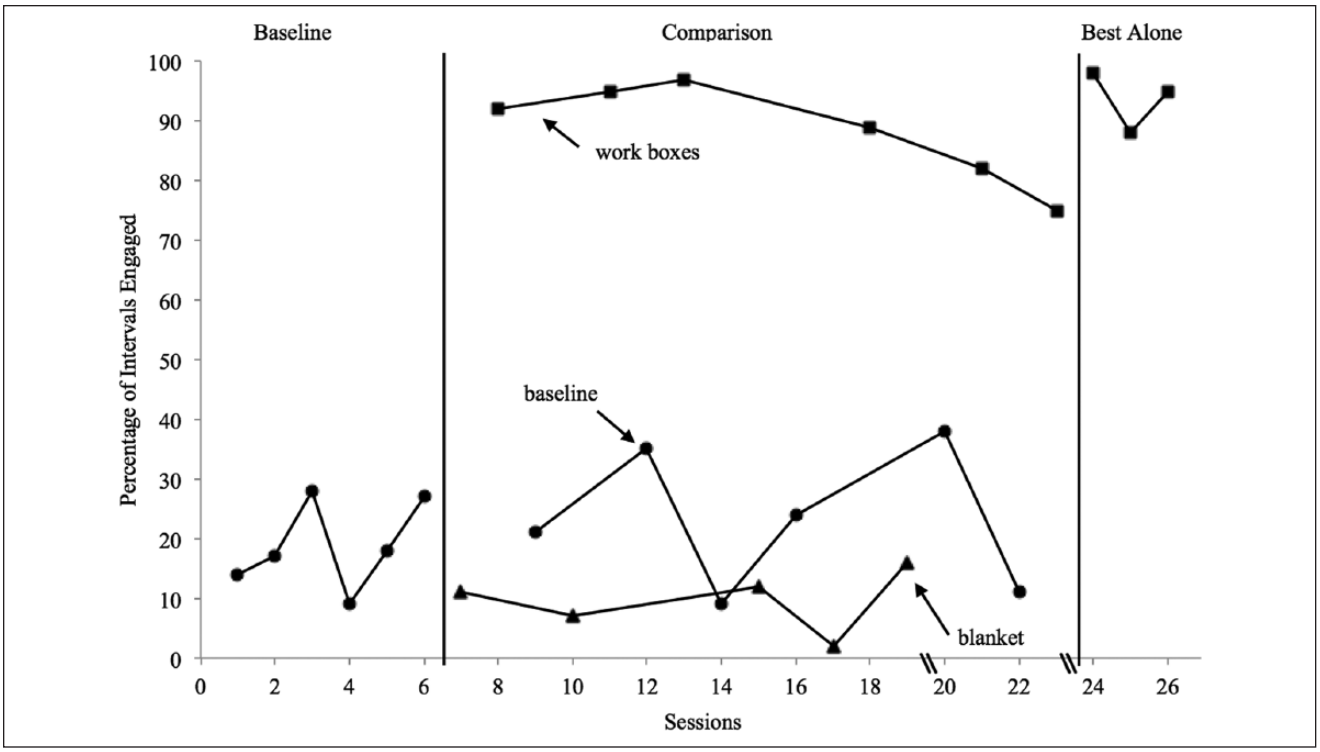


Figure 1. Percentage of intervals engaged during baseline, weighted blanket, and structured work box conditions.
Note. The first hash mark indicates a break due to student illness and Thanksgiving Break (14 weekdays); the second indicates Winter Break for the school (20 weekdays).

potential interactions—for example, it would have allowed us to determine whether the blanket resulted in similarly undifferentiated outcomes in contexts with higher engagement (i.e., if the SWB + weighted blanket condition was similar to SWB alone). Despite this limitation, the lack of differentiation between baseline and weighted blanket conditions remains clear.

Results should be considered in light of some additional limitations including the use of a single participant; the effectiveness of weighted blanket interventions should be evaluated with additional students in future research. In addition, Anwar was not learning new information in the math circle time, which provided whole-group opportunities to practice learned skills; the SWB intervention fit well with this purpose, but might not be appropriate for other types of activities. Although intervention was implemented by a researcher, the classroom general education teacher reported continued use of the SWB intervention after study completion, suggesting the intervention is feasible for practitioner implementation. Finally, although not experimentally evaluated, the teacher also reported the use of a weighted blanket was not added to the student's IEP given the results of this study. Despite their popular use, results of this study suggest weighted blankets may not result in improved outcomes for students with ASD. Additional investigations of low effort, effective interventions like SWB are needed to continue to identify ways to improve engagement for students with ASD engaging in stereotypy during typical classroom routines.

Authors' Note

Since the time this research was completed, author Kathleen N. Zimmerman has changed affiliation. She is now with the University of Kansas.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

References

Ayres, K., & Ledford, J. R. (2014). Dependent measures and measurement systems. In D. L. Gast & J. R. Ledford (Eds.), *Single case research methodology* (pp. 124–153). New York, NY: Routledge.

Barlow, D. H., & Hayes, S. C. (1979). Alternating treatments design: One strategy for comparing the effects of two

treatments in a single subject. *Journal of Applied Behavior Analysis*, *12*, 199–210.

Barton, E. E., Reichow, B., Schnitz, A., Smith, I. C., & Sherlock, D. (2015). A systematic review of sensory-based treatments for children with disabilities. *Research in Developmental Disabilities*, *37*, 64–80.

Bennett, K., Reichow, B., & Wolery, M. (2011). Effects of structured teaching on the behavior of young children with disabilities. *Focus on Autism and Other Developmental Disabilities*, *26*, 143–152.

Goin-Kochel, R. P., Mackintosh, V. H., & Myers, B. J. (2009). Parental reports on the efficacy of treatments and therapies for their children with autism spectrum disorders. *Research in Autism Spectrum Disorders*, *3*, 528–537.

Gresham, F., & Elliott, S. N. (2008). *Social Skills Improvement System Rating Scales*. Minneapolis, MN: Pearson Education.

Gringras, P., Green, D., Wright, B., Rush, C., Sparrowhawk, M., Pratt, K., . . . Wiggs, L. (2014). Weighted blankets and sleep in autistic children—A randomized controlled trial. *Pediatrics*, *134*, 298–306.

Hanson, E., Kalish, L. A., Bunce, E., Curtis, C., McDaniel, S., Ware, J., & Petry, J. (2007). Use of complementary and alternative medicine among children diagnosed with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, *37*, 628–636.

Losinski, M., Cook, K., Hirsch, S., & Sanders, S. (2017). The effects of deep pressure therapies and antecedent exercise on stereotypical behaviors of students with autism spectrum disorders. *Behavioral Disorders*, *42*, 196–208.

Morrison, K., & Rosales-Ruiz, J. (1997). The effect of object preferences on task performance and stereotypy in a child with autism. *Research in Developmental Disabilities*, *18*, 127–137.

Rogers, S. J., Hepburn, S., & Wehner, E. (2003). Parent reports of sensory symptoms in toddlers with autism and those with other developmental disorders. *Journal of Autism and Developmental Disorders*, *33*, 631–642.

Tapp, J. (2003). ProcoderDV. Nashville, TN: Vanderbilt Kennedy Center.

Wong, C., Odom, S. L., Hume, K., Cox, A. W., Fettig, A., Kucharczyk, S., . . . Schultz, T. R. (2014). *Evidence-based practices for children, youth, and young adults with autism spectrum disorder*. Retrieved from http://cidd.unc.edu/Registry/Research/Docs/31.pdf

Zane, T., Davis, C., & Rosswurm, M. (2008). The cost of fad treatments in autism. *Journal of Early and Intensive Behavior Intervention*, *5*, 44–51.